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Routing at the Network Layer

(a) Static Routing Protocol Configuration Using Cisco Packet Tracer.

Aim:

To Configure static routing b/w different networks using Cisco Packet Tracer and verify packet

Alg/Procedure:

1. Open Cisco packet Tracer and places two routers, Switches, and PCs.
2. Connect the PCs, to their respective Switches and routers.
3. Assign IP address and Subnet masks to the PCs and router interfaces.
4. Configure the IP address of the router interfaces and enable them.
5. Configure the IP addresses of the inter-router links.
6. Add Static routes on each router using the IP route command.
7. Verify the routing table [Router# show ip route]
8. Use ping to test communication b/w PCs on different networks.
9. Observe packet transmission in Simulation mode

Example

Router (config)# ip route <destination - network> <subnet mask> <next-hop-IP>

Command prompt / Router CLI:

Router > enable

Router # configure terminal

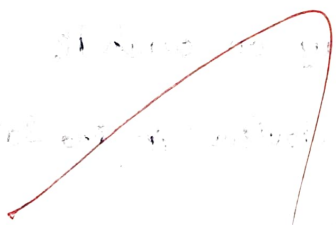
Router (config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2

Router (config)# end

Router# show ip route

Sample output:

S 192.168.2.0/24 [1/0] via 10.0.0.2



(b) RIP Configuration Using Cisco Packet Tracer.

Aim:

= To Configure the Routing Information Protocol (RIP) and verify dynamic routing between different networks using Cisco Packet Tracer.

Algorithm/procedure:

1. Open Cisco Packet Tracer and create a topology with multiple routers and networks.
2. Connect the routers and end devices using suitable cables.
3. Assign IP addresses and subnet masks to all devices and router interfaces.
4. Enable the router interfaces using the no shutdown command.
5. Enter RIP configuration mode on each router.
6. Enable RIPv2 and advertise the directly connected networks.
7. Configure the appropriate network statements on every router.
8. Verify the routing table using show ip route.
9. Confirm that routes learned through RIP appear with the code R
10. Test connectivity b/w PCs in different networks using Ping.

RIP Routing

```
Router> enable
Router# configure terminal
Router(Config)# router rip
Router(Config-router)# version 2
Router(Config-router)# network 192.168.1.0
Router(Config-router)# network 192.168.2.0
Router(Config-router)# end
Router# show ip route
```

Sample output

```
R  192.168.2.0/24 [120/1] via 10.0.0.2
```



Result:

RIP was configured successfully. The routers exchanged routing information dynamically, learned remote network routes, and forwarded packets b/w the networks.